

# Benchmark Radar v0.9.0

From daily collection to benchmark search and score history

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|                                                            |                                                                      |                                                               |                                                     |
|------------------------------------------------------------|----------------------------------------------------------------------|---------------------------------------------------------------|-----------------------------------------------------|
| <b>7,540</b><br>source observations<br>across 37 snapshots | <b>4,537</b><br>unique artifacts in the<br>cumulative evidence graph | <b>1,242</b><br>searchable entries<br>across 4 search sources | <b>37</b><br>public collection<br>sources monitored |
|------------------------------------------------------------|----------------------------------------------------------------------|---------------------------------------------------------------|-----------------------------------------------------|

**Benchmark Radar brings daily discoveries, catalog search, vendor reporting, and sourced model scores into one place. Each layer keeps its own count because each records a different kind of evidence.**

## Executive findings

- Search covers 1,242 entries. Three external catalogs supply 1,173 rows. The curated score archive adds 69 benchmark tracks.
- The daily corpus holds 7,540 observations for 4,537 artifacts. Primary or structured sources supplied 7,523 observations, and each record keeps its source URL and retrieval metadata.
- Frozen v0.9.0 population: 36 model reports cover 94 curated benchmarks. Those cutoff counts remain the deposited report baseline. The issue #456 addendum separately audits 37 reviewed documents, 12 organization labels, and 110 benchmark IDs pinned at commit 98c8cf6.
- Identity and task classification need more work. Forty-one artifacts link across multiple sources. KW-Bench has produced 4,129 tracks, with classification still empty in v0.9.0.

## Abstract

Benchmark Radar collects benchmark work from 37 public sources and publishes it through a dashboard, RSS, JSON, and an offline client. This report checks the full path from collection to search. It covers source health, exact-identifier matching, the ranking rubric, the 1,242-entry search index, model-card adoption, score histories, and the reports already in the repository. The system has strong provenance and reproducible builds. Cross-source matching, protocol capture, semantic search, and KW-Bench classification remain active work.

# 1. Product tour

The README leads with the user task: find a benchmark in seconds, then inspect model-card adoption and score movement. The dashboard and RSS feed serve readers. JSON, the local CLI, and the local HTTP API support reproducible research.

Benchmark Radar's daily-intelligence logic was inspired by BuilderPulse, an AI-powered daily intelligence project for indie hackers and builders that answers 20 questions from 10+ sources every morning [8]. Benchmark Radar adapts that multi-source briefing pattern to benchmark discovery, normalized catalog search, and score history.

Search every benchmark

Search benchmarks, tasks, domains...

1,242 benchmarks · 4 sources

## 1.1 Search coverage

| Search layer         | Rows  | Best field                                    | Score data                                   |
|----------------------|-------|-----------------------------------------------|----------------------------------------------|
| LLM Stats            | 687   | Benchmark names and descriptions              | 5,544 rows; evaluation setup is sparse       |
| OpenCompass Hub      | 461   | Paper, repository, release, and dataset links | Historical leaderboards stay source-labelled |
| Artificial Analysis  | 25    | Current commercial evaluation catalog         | 7,050 third-party rows                       |
| Curated score tracks | 69    | Scores read from primary model reports        | 285 rows with instrument and protocol fields |
| Search box           | 1,242 | All four layers in one index                  | Source label shown on every result           |

The UI adds 1,173 external rows and 69 curated score tracks. It keeps source records separate, so the same benchmark name can appear in more than one catalog. The label “4 sources” refers to these four search layers. The daily collector uses 37 public endpoints.

## 1.2 Dashboard and local tools

| Surface      | Reader question                                 | Evaluation                                           |
|--------------|-------------------------------------------------|------------------------------------------------------|
| Today        | What appeared or changed?                       | Ranks new records and links the source evidence.     |
| Search       | Which records match this name, task, or domain? | Ranks exact names, phrases, and field-token matches. |
| Leaderboard  | Which benchmarks do labs report?                | Counts benchmark mentions across vendor reports.     |
| Scores       | Which printed values can be connected?          | Matching instrument + protocol creates a series.     |
| Trends / map | Which topics and sources recur?                 | Coverage signatures gate comparisons.                |
| CLI + HTTP   | Can an analyst reproduce search offline?        | One QueryService and stable JSON contract.           |

## 2. Pipeline evaluation



*The same snapshots and code produce the same derived files.*

### 2.1 Collection and health

Each run queries enabled connectors inside a 48-hour window, records counts and errors, drops future-dated rows, then requires the configured core sources to be healthy. arXiv, Hugging Face, and GitHub are required. On the cutoff run all three were healthy; Semantic Scholar returned HTTP 429 and Brave had no API key; OpenReview was healthy but empty.

The cutoff run fetched 826 rows, deduplicated them to 783, classified 273 as eligible, and recommended 97. Two collections ran that day. Their merged page contains 528 records, including 186 recommendations.

### 2.2 Normalization, identity, and retention

- Stable fields include source ID, URL, title, timestamps, authors or organizations when supplied, parser version, retrieval time, and a SHA-256 payload fingerprint. The publisher omits raw responses and credentials.
- The resolver merges observations that share a DOI, arXiv ID, OpenReview ID, GitHub repository, or Hugging Face artifact. A similar title leaves two records in place until a reviewer confirms the match.
- The corpus includes eligible records below the recommendation threshold. The recommended tag controls display priority.

### 2.3 Interpretation, publication, and query

Priority combines relevance (35%), evidence (20%), recency (20%), and adoption (25%) on a 0–100 scale. Scoring version 5 ships with the data. Generators replay validated snapshots into the cumulative graph, normalize external catalogs, classify KW-Bench tracks, and package a checksummed release. Installed clients switch to a new data version after checksum and schema validation. Search reads the active local version.

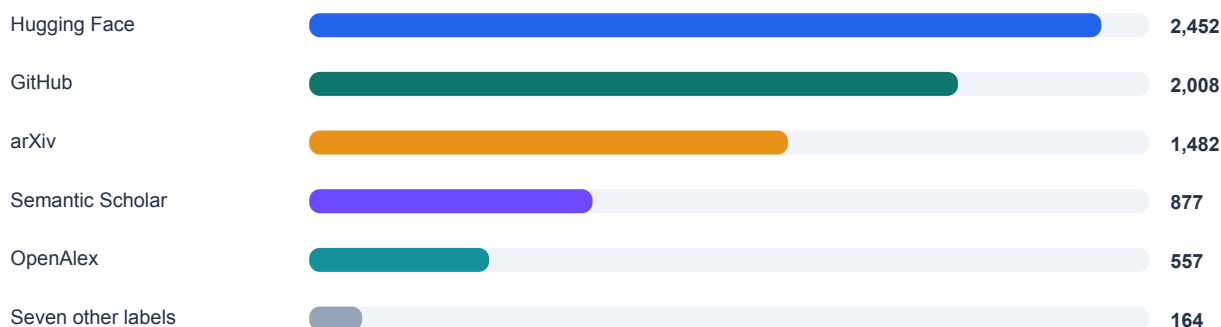
| Area                   | Working today                                                              | Current gap                                        |
|------------------------|----------------------------------------------------------------------------|----------------------------------------------------|
| <b>Reproducibility</b> | Snapshots, schemas, parser versions, hashes, deterministic rebuilds.       | Four early snapshots are simulated.                |
| <b>Reliability</b>     | Required-source gate; optional failures visible; atomic client activation. | Rate limits and secrets change realized coverage.  |
| <b>Identity</b>        | Exact-anchor merges and reviewed external groups.                          | Forty-one artifacts have more than one source.     |
| <b>Interfaces</b>      | CLI and HTTP share QueryService and JSON.                                  | Local token matching misses some paraphrases.      |
| <b>Verification</b>    | Clean-worktree rebuild plus a passing full CI suite.                       | Source coverage still depends on public endpoints. |

### 3. Counts and units

The dashboard publishes several counts because it tracks several units. Use the unit and cutoff beside the number.

| Unit / product                      | Count | Definition                                                | Use it for                                   |
|-------------------------------------|-------|-----------------------------------------------------------|----------------------------------------------|
| Snapshot                            | 37    | 33 observed, 4 simulated.                                 | The dated history available for replay.      |
| Source observation                  | 7,540 | Persisted source record.                                  | Evidence volume in cumulative replay.        |
| Unique artifact                     | 4,537 | Exact-ID-resolved paper, repo, dataset, release, or page. | Distinct artifacts observed.                 |
| External catalog row                | 1,173 | One row from three catalog sources.                       | Broad, source-labelled catalog search.       |
| Searchable entry                    | 1,242 | 1,173 external + 69 curated score tracks.                 | Current web-search reach.                    |
| Adoption benchmark (frozen v0.9.0)  | 94    | Canonical identity in curated registry.                   | Model-card adoption and missing adoption.    |
| Model-card document (frozen v0.9.0) | 36    | Curated reports from 11 organizations.                    | Documents read for mentions.                 |
| Curated score                       | 285   | Numeric result from a cited document; 69 tracks.          | Score histories grouped by evaluation setup. |
| Model registry                      | 861   | Models across curated and crawled layers; 19 in both.     | Finding a model across both data layers.     |

#### 3.1 Cumulative source composition



7,540 cumulative source observations through 29 August 2026

Five normalized source labels supply 7,376 of 7,540 observations (97.8%). The other endpoints add discovery routes and publish their own yield and health status.

#### 3.2 Adoption and score findings

The issue #456 sensitivity audit rebuilds vendor-reporting counts from the reviewed registry rather than repeating the frozen PDF prose. Its primary full-history rule finds 16 canonical benchmark IDs reported by at least six organizations, while the pre-registered document, time-window, family, and missing-report alternatives produce different boundaries. Section 6.1 reports the result and limitations. The score archive remains separate: it finds eight bounded metrics with five points of headroom or less, and six benchmarks gained model-card mentions after their last readable score.

## 4. Source inventory and health

The daily collector reads 12 discovery connectors, 24 first-party feeds, and Hacker News. The search box draws from four catalog layers.

### 4.1 Direct discovery connectors (12)

| Connector            | Role                                  | Cutoff run            | Core? |
|----------------------|---------------------------------------|-----------------------|-------|
| arXiv                | Primary papers via cs.AI/CL/CV/SE RSS | Healthy · 23          | Yes   |
| Hugging Face Hub     | Datasets and Spaces                   | Healthy · 126         | Yes   |
| GitHub Search        | Code and artifacts                    | Healthy · 300; at cap | Yes   |
| GitHub Organizations | Reviewed organization repos           | Healthy · 15          | No    |
| Hugging Face Papers  | Community-surfaced papers             | Healthy · 23          | No    |
| Kaggle Datasets      | Public benchmark datasets             | Healthy · 28          | No    |
| Zenodo               | DOI-bearing artifacts                 | Healthy · 77          | No    |
| OpenReview           | Conference submissions                | Healthy · 0           | No    |
| Semantic Scholar     | Structured scholarly discovery        | Failed · HTTP 429     | No    |
| GitHub Releases      | Curated first-party releases          | Healthy · 3           | No    |
| OpenAlex             | Scholarly discovery                   | Healthy · 228         | No    |
| Brave Search         | Web and official domains              | Unavailable · no key  | No    |

### 4.2 First-party feeds (24)

| Feeds 1–12                                                                                                                                                                                                                                   | Feeds 13–24                                                                                                                                               |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Meituan Engineering<br>OpenAI News<br>Google AI<br>Google DeepMind<br>Google Research<br>Apple Machine Learning Research<br>AWS Machine Learning<br>Hugging Face Blog<br>Microsoft Research<br>NVIDIA AI Blog<br>Mistral AI<br>Meta Research | Ai2<br>Together AI<br>Sakana AI<br>Qwen<br>Ollama<br>Stability AI<br>Nomic AI<br>Replicate<br>NVIDIA Developer<br>IBM Research<br>Databricks<br>LangChain |

The feed collector yielded three relevant records. Some healthy feeds produced zero matches after relevance filtering. Domain-constrained searches cover organizations that lack a verified feed. The collector excludes third-party mirrors.

### 4.3 Public attention (1)

The Hacker News collector uses the public Algolia API. It found 12 records on the cutoff run and places them in a separate attention feed. The priority calculation uses evidence sources only.

## 5. Data quality and report history

| Finding            | Evidence                                                   | Interpretation                                                     |
|--------------------|------------------------------------------------------------|--------------------------------------------------------------------|
| High provenance    | 7,523 / 7,540 primary or structured (99.77%).              | Readers can open the upstream record for almost every observation. |
| Low corroboration  | 41 / 4,537 artifacts have >1 normalized source (0.90%).    | Exact identity prevents bad joins but leaves plausible duplicates. |
| Uneven yield       | Five source labels provide 97.8% of observations.          | Caps or outages can change apparent topic mix.                     |
| Simulation         | 23–26 July are simulated snapshots.                        | Use them to test replay and UI history.                            |
| Classification gap | KW-Bench: 4,129 tracks, 0 classified.                      | The task-capability view has no classifications yet.               |
| Protocol sparsity  | 12,594 aggregator scores stay outside curated progression. | Comparison needs shots, harness, tools, attempts, and date.        |
| Lexical retrieval  | Field tokens drive rank; no semantic reranker.             | Reproducible, but paraphrases can be missed.                       |

### 5.1 Existing reports

| Document                 | Use today                                                                           | Date limit                                                                    |
|--------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| Landscape report, 31 Jul | A dated study of 791 sightings, 645 artifacts, and 78 agentic-evaluation artifacts. | Its totals stop on 31 July, before the external catalog and newer connectors. |
| Source probes, 27 Aug    | Verified HF Papers, Kaggle, Spaces, and Zenodo endpoints.                           | The probes record one check on 27 August.                                     |
| External catalog audit   | OpenCompass identity fields and the score-heavy peer catalogs.                      | The audit keeps same-name records separate until identity review.             |
| Daily report / briefing  | Triage with citations and source health.                                            | Each edition covers one collection window.                                    |

### 5.2 Reading the evidence

- You can trace a new or updated record to its source, inspect why the taxonomy matched it, and find the vendor document behind each curated benchmark mention.
- For score comparisons, match the instrument and protocol fields. For trend comparisons, use days with the same connector coverage and report limit. Market-size estimates need a separate benchmark-family census.

## 6. Findings from the current data

Benchmark Radar turns papers, repositories, datasets, and model reports into records you can search, filter, download, and query offline. Three results stand out in the current data.

### 6.1 A recurring reporting group with a definition-sensitive boundary

| Definition                       | Observed threshold set                                                     |
|----------------------------------|----------------------------------------------------------------------------|
| Full reviewed history            | 16 canonical IDs · 37 documents · 12 organizations                         |
| Trailing 365 days                | 6 IDs · same 6-organization threshold                                      |
| Latest document per organization | 3 IDs · 12 documents                                                       |
| Reviewed family projection       | 13 resolved identities · 5 families + 8 singletons · score tracks unmerged |

The original eight-item statement does not reproduce from its stated rule. Across all 37 reviewed documents from 12 organization labels, 16 canonical benchmark IDs—not eight—appear in documents from at least 6 organizations. The eight names printed in the prior draft are a strict subset of that threshold set and match a ranking truncation, not a complete threshold-defined core.

The boundary changes under the pre-registered alternatives. The trailing 365-day window contains 6 IDs at the same threshold; one latest document per organization contains 3; model-card documents alone contain 4; and the explicit reviewed-family projection contains 13 resolved identities: 5 explicit families and 8 singleton canonical IDs. These results support a narrower claim that a recurring reporting group exists in the reviewed sample. They do not support an exact eight-benchmark boundary or a field-wide statement that vendor attention has converged.

Method and limits: Junkai Wang / @JunkaiWang-TheoPhy contributed the issue #456 audit (<https://github.com/ktwu01/benchmark-radar/issues/456>) at source commit 98c8cf6b5d1d69c66d438ea9f92242b2205c9ae and cutoff 2026-08-31. The exact-membership median Jaccard is 0.4688 including the baseline self-comparison (excluding it: 0.3750; required 0.80), so the threshold set is not retained as an invariant. Counts are rebuilt from data/model\_cards.yml as binary organization-by-benchmark mention edges; documents, models, benchmark IDs, families, and score tracks remain separate. The sensitivity grid varies thresholds, document types, latest-report selection, time windows, reviewed families, and missing-report stress tests. The registry is a reviewed convenience sample rather than a census: a not-observed cell can mean an unread or missing report and is not evidence of vendor omission. Every aggregate links back to model-card IDs and URLs in the issue #456 machine-readable audit tables [9-10].

### 6.2 All eight raw near-ceiling readings come from one-date setups

All eight raw best readings leave five points of headroom or less, and every exact raw-best instrument+protocol setup appears on only one date. Four benchmarks have a different setup repeated across dates. Only HMMT's closest repeated setup is also within five points, and all four repeated setups are two-date, single-organization pairs. The remaining four benchmarks have no repeated setup. The evidence supports protocol-specific headroom claims only.

| Benchmark          | Raw gap | Same setup | Repeat gap | Claim decision | Note              |
|--------------------|---------|------------|------------|----------------|-------------------|
| AIME               | 0.8     | 1 date     | 20.2       | replace        | 2 dates · 1 org   |
| Arena-Hard         | 4.4     | 1 date     | 7.7        | replace        | 2 dates · 1 org   |
| DeepSearchQA       | 5       | 1 date     | n/a        | qualify        | no repeated setup |
| HMMT               | 3.1     | 1 date     | 4.8        | qualify        | 2 dates · 1 org   |
| MATH-500           | 2       | 1 date     | n/a        | qualify        | no repeated setup |
| MathVision         | 2.2     | 1 date     | n/a        | qualify        | no repeated setup |
| SWE-bench Verified | 4       | 1 date     | 19.4       | replace        | 2 dates · 1 org   |
| tau2-bench         | 0.7     | 1 date     | n/a        | qualify        | no repeated setup |

| Threshold | Raw readings | Repeated setups   |
|-----------|--------------|-------------------|
| <=5       | 8 / 8        | 1 / 4 (4 unknown) |

| Threshold     | Raw readings | Repeated setups   |
|---------------|--------------|-------------------|
| <b>&lt;=3</b> | 4 / 8        | 0 / 4 (4 unknown) |
| <b>&lt;=2</b> | 3 / 8        | 0 / 4 (4 unknown) |

Repeated-setup sensitivity is 1 of 4 assessable benchmarks at five points; using 8 as the denominator would count four unknowns as negatives. No raw-best setup qualifies as repeat-controlled. The machine-readable audit beside the report source records every protocol stratum, score ID, exclusion, counterexample, and exact decision wording.

## Per-benchmark claim wording

| Benchmark                 | Exact report wording                                                                                                                                                                                                                                                    |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>AIME</b>               | AIME's 99.2 raw best leaves 0.8 points of headroom under the AIME 2026, temp 1.0, top_p 0.95, max gen 163840 tokens, GPT-5.5 (medium) judge setup, but that setup appears on one date. The closest repeated setup leaves 20.2 points, outside the five-point threshold. |
| <b>Arena-Hard</b>         | Arena-Hard's 95.6 raw best leaves 4.4 points of headroom under the thinking mode setup, but that setup appears on one date. The closest repeated setup leaves 7.7 points, outside the five-point threshold.                                                             |
| <b>DeepSearchQA</b>       | DeepSearchQA's 95 result leaves 5 points of headroom under the F1, reasoning=max setup. No setup spans two dates, so the archive supports only this protocol-specific reading.                                                                                          |
| <b>HMMT</b>               | HMMT's raw best is an isolated 96.9 result. A separate think max, pass@1 setup spans 2 dates from 1 organization and leaves 4.8 points of headroom. This supports a protocol-specific paired comparison.                                                                |
| <b>MATH-500</b>           | MATH-500's 98 result leaves 2 points of headroom under the thinking mode setup. No setup spans two dates, so the archive supports only this protocol-specific reading.                                                                                                  |
| <b>MathVision</b>         | MathVision's 97.8 result leaves 2.2 points of headroom under the reasoning=max, with Python setup. No setup spans two dates, so the archive supports only this protocol-specific reading.                                                                               |
| <b>SWE-bench Verified</b> | SWE-bench Verified's 96 raw best leaves 4 points of headroom under the adaptive thinking, max effort, averaged over 5 trials setup, but that setup appears on one date. The closest repeated setup leaves 19.4 points, outside the five-point threshold.                |
| <b>tau2-bench</b>         | tau2-bench's 99.3 result leaves 0.7 points of headroom under the Thinking (High), Telecom setup. No setup spans two dates, so the archive supports only this protocol-specific reading.                                                                                 |

## 6.3 Broad search, deeper curation

The external catalog holds 1,173 rows, more than twelve times the 94-benchmark adoption registry. Use catalog search to find candidates. The curated registry adds the model reports, organizations, instruments, and protocols needed for comparison.

## 6.4 What to build next

| Priority | Measurement work                                                    | Result                                                   |
|----------|---------------------------------------------------------------------|----------------------------------------------------------|
| <b>1</b> | Show the count unit beside every UI total and export.               | Readers can see which population each number counts.     |
| <b>2</b> | Resolve high-value cross-source identities with anchors and review. | More records gain paper, code, and dataset links.        |
| <b>3</b> | Complete KW-Bench extraction and reviewed coverage.                 | Turns 4,129 unclassified tracks into a task map.         |
| <b>4</b> | Expand protocol capture from newer model cards.                     | Closes the mention-versus-score recency gap.             |
| <b>5</b> | Add optional semantic retrieval while retaining lexical reasons.    | Finds paraphrases while retaining visible match reasons. |



## 6.5 Worked real use case: prior-art check for a new evaluation

Jiayu Wang, a researcher working on agent evaluation, used Benchmark Radar to decide whether a proposed new evaluation would duplicate existing work. The check decides whether the design is still novel, and it used to be slow: comparing a candidate against the field required long manual searches, and completeness was hard to guarantee. The case ran during August 2026 with a concrete task: survey recent work on credit assignment in agentic training, keeping small Qwen-series baselines as a reproducibility constraint.

The author gave the task to a coding agent together with the public consumer prompt for Benchmark Radar. The agent installed the CLI and the benchmark-radar Skill, initialized the local corpus with benchmark-radar init, and queried candidate records with benchmark-radar search.

```
what can i do with benchmark-radar cli?

+ Thought: 328ms

$ benchmark-radar --help
usage: benchmark-radar [-h] [--author-output AUTHOR_OUTPUT]
                        [--author-contacts AUTHOR_CONTACTS]
                        [--author-repo-limit AUTHOR_REPO_LIMIT]
                        [--author-contributors AUTHOR_CONTRIBUTORS]
                        [--author-emails] [--config CONFIG] [--output OUTPUT]
                        [--json-output JSON_OUTPUT]
                        [--social-output SOCIAL_OUTPUT] [--items ITEMS]
                        [--repo REPO] [--channels CHANNELS]
                        [--existing-body EXISTING_BODY] [--since SINCE]
                        [--until UNTIL] [--snapshot-dir SNAPSHOT_DIR]

-
Click to expand

+ Thought: 531ms

The CLI has two groups:

Consumer (query) commands – for searching and inspecting local benchmark data:
- search "query" --scope catalog|radar|all --json – discover benchmarks (catalog = verified records, radar = recent unverified evidence like papers/repos/datasets)
- show "key-or-slug" --json – inspect one record in detail
- recent --json – newest Radar evidence
- status --json – local data health/provenance
- init / sync – set up or refresh the local data
- serve --host 127.0.0.1 --port 8765 – local HTTP interface for the same queries

Maintainer commands – for operating the radar pipeline (not for ordinary use): run, rebuild, backfill, migrate, rescore, simulate-history, export, classify, authors, social, normalize-external, build-data-release.

For your local benchmark search use case, you'll mostly need search, show, recent, status, and serve.

Build - DeepSeek V4 Flash DeepSeek
..... esc Interrupt 150.1K (15%) · $0.83 ctrlap commands
```

Figure 1. A coding agent follows the consumer setup prompt, installs the Benchmark Radar CLI and Skill, and runs local queries.

Radar links one artifact across papers, code, releases, and datasets. The agent could therefore see at a glance whether a candidate was announced as a paper with no released code, or shipped code without its dataset.

```
    "rank": 18,
    "recommended": false,
    "score": 18.75,
    "slug": null,
    "snapshot_date": "2026-09-01",
    "source": "Hugging Face Papers",
    "source_id": "2608.28476",
    "updated_at": "2026-08-28T00:00:00+00:00",
    "url": "https://huggingface.co/papers/2608.28476"
  },
  {
    "categories": [
      "benchmark"
    ],
    "description": "LTABC-KM is a long-tail-aware adaptive artificial bee colony-K-means for unsupervised image clustering. It leverages density-aware coresets and graph-propagation assignment to recover minority tail modes without labels, achieving significant Macro-F1, NMI and ARI improvements on imbalanced benchmarks with favorable quality-computation trade-offs.",
    "has_dataset": false,
    "has_paper": false,
    "has_repo": true,
    "has_size": false,
    "key": "radar:github:Lutra11/LTABC-KM",
    "kind": "radar",
    "languages": [],
    "match": {
      "idf_coverage": 0.4391,
      "matched_fields": [
        "description"
      ],
      "matched_tokens": [
        "assignment"
      ],
      "missing_tokens": [
        "credit",
        "training"
      ],
      "phrase_fields": [],
      "query_coverage": 0.3333,
      "ranking_algorithm": "bm25f_v3",
      "reason": "1 of 3 unique query tokens matched across fields; IDF coverage 0.44",
    }
  }
]
```

Build - DeepSeek V4 Flash DeepSeek

D:\projects\radar-test 187.3K (19%) · \$0.04 ctrl+p commands

Figure 2. Radar consolidates the sources and status of one artifact.

```
    "source": "GitHub",
    "source_id": "mdkamranalam/credixa",
    "updated_at": null,
    "url": "https://github.com/mdkamranalam/credixa"
  },
  {
    "categories": [
      "benchmark"
    ],
    "description": "Long-horizon agentic tasks require large language models (LLMs) to iteratively retrieve, integrate, and maintain dispersed information across multi-turn interactions, but preserving all interaction histories leads to a continuously growing working context. Recent proactive context management methods allow models to edit their own working context with specialized tools, yet they still face three key limitations: (1) a limited toolset restricted to search, deletion, and summarization, with no support for global planning, long-term memory, and adaptive compression; (2) inefficient exploration that treats context management actions uniformly despite their heterogeneous impacts on final outcomes; and (3) coarse-grained credit assignment that assigns the final trajectory-level reward to all intermediate context editing actions during RL. To bridge these gaps, we introduce ContextPilot, a proactive context management framework for long-horizon agentic reasoning. Our approach systematically augments the toolset with planning, long-term memory, and soft context offloading tools. We further propose an RL method tailored for context management, which uses context and entropy variation to identify critical editing decisions for branch sampling and estimates action-level advantages from all branched trajectories that pass through the corresponding context editing action. Experiments on long-context QA and deep search tasks show that ContextPilot achieves stronger performance with a more compact working context, consistently outperforming existing baselines across various base models and benchmarks. Code is available at https://github.com/Tencent/ContextPilot.",
    "has_dataset": false,
    "has_paper": true,
    "has_repo": true,
    "has_size": false,
    "key": "radar:hugging face papers:2608.28476",
    "kind": "radar",
    "languages": [],
    "match": {
      "idf_coverage": 0.8392,
      "matched_fields": [
        "description"
      ],
      "matched_tokens": [
        "assignment",
        "credit"
      ],
    }
  }
]
```

Build - DeepSeek V4 Flash DeepSeek

D:\projects\radar-test 187.3K (19%) · \$0.04 ctrl+p commands

Figure 3. A companion record in which code is public but the dataset is not yet released.

Radar search is deterministic lexical matching, so the agent also ran its own web search and cross-checked the two candidate sets before accepting a record. This double pass keeps a differently worded version of the same idea from being missed.

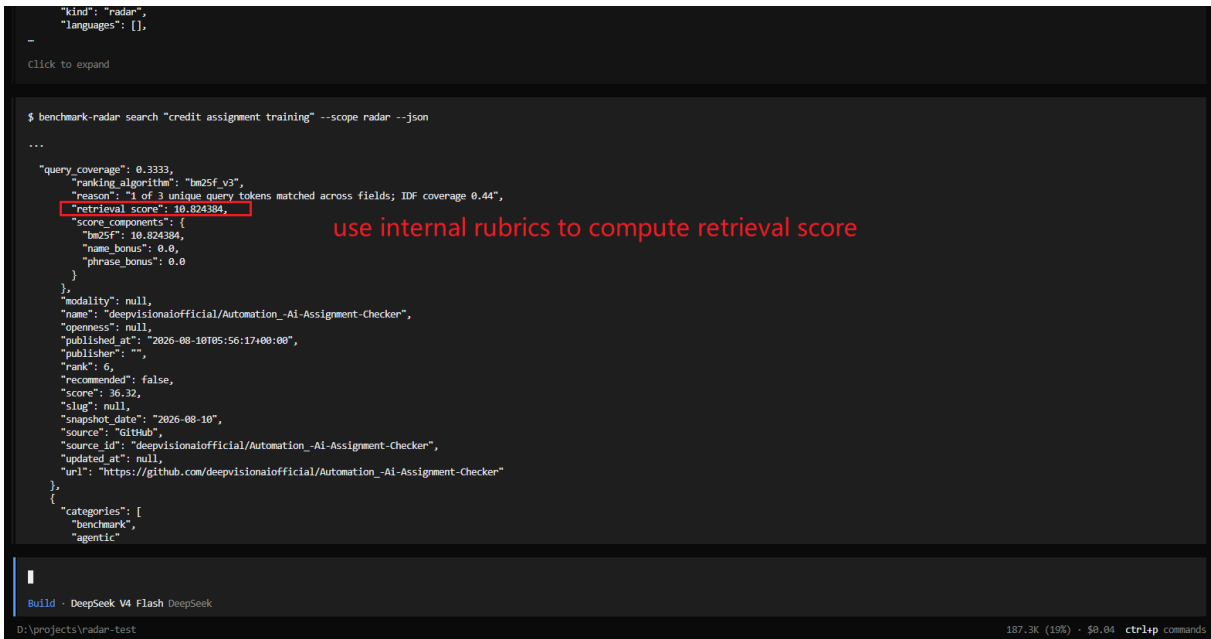


Figure 4. Cross-checking Radar candidates against the agent’s own web search before accepting a record.

The session ended with a focused summary table of related work that the author judged complete enough to act on.

| # | Work                                                                               | Published  | Qwen base model(s)                                                         | Credit-assignment focus                                                                                              | Agentic benchmarks used                                                | Link                                                                                                                                                                            |
|---|------------------------------------------------------------------------------------|------------|----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | SRPO: Self-Reflective Policy Optimization for Long-Horizon Reasoning               | 2026-08-25 | Qwen3-8B (also Qwen3-1.7B / 32B scaling)                                   | Reflection-conditioned dense token-level signals convert sparse terminal reward into training signal                 | AIIME'24, WebShop, ALFWorld, SWE-Bench-Lite                            | <a href="https://arxiv.org/abs/2608.23493">https://arxiv.org/abs/2608.23493</a> · <a href="https://github.com/Galleons2029/SRPO">https://github.com/Galleons2029/SRPO</a>       |
| 2 | ContextPilot: Teaching Agents for Proactive Context Management via Fine-grained RL | 2026-08-28 | Qwen3-8B, Qwen3-14B                                                        | Coarse-grained → action-level credit assignment over branched trajectories during agentic RL                         | InfBench (InfiniteBench), NovelQA, LongMemEval, BrowseComp+            | <a href="https://arxiv.org/abs/2608.28476">https://arxiv.org/abs/2608.28476</a> · <a href="https://github.com/Tencent/ContextPilot">https://github.com/Tencent/ContextPilot</a> |
| 3 | SkillGate: Training In-Policy Skill Selection in Long-Horizon Agents               | 2026-08-21 | Qwen3.5-9B (SFT checkpoint, RL init)                                       | "Selector credit starvation": partitions token support into two credit channels (outcome vs. action-local advantage) | 5 agentic benchmarks incl. SkillsBench, SWE (16-candidate skill slate) | <a href="https://arxiv.org/abs/2608.18852">https://arxiv.org/abs/2608.18852</a>                                                                                                 |
| 4 | CIPO: Contextual Information Policy Optimization for Search Agents                 | 2026-08-06 | Qwen2.5-3B-Instruct, Qwen2.5-7B-Instruct                                   | Dense turn-level credit (EALR) to evidence-using reasoning actions, combined with outcome reward                     | HotpotQA, 2WikiMultiHopQA, MuSiQue, Bamboogle + 3 OOD                  | <a href="https://arxiv.org/abs/2608.06128">https://arxiv.org/abs/2608.06128</a>                                                                                                 |
| 5 | MoRSE: Task-Oriented Multi-Agent System with Mixture of Role-Subtask Experts       | 2026-08-10 | Qwen3-4B-Instruct (one of three backbones; also Llama-3.1-8B, Gemma-4-31B) | Hierarchical GRPO with two-layer credit assignment (isolates expert vs. routing quality)                             | Code-generation benchmarks (multi-agent), held-out task domains        | <a href="https://arxiv.org/abs/2608.09251">https://arxiv.org/abs/2608.09251</a>                                                                                                 |

Figure 5. Summary table of recent work on credit assignment in agentic training assembled during the session.

The savings are easiest to measure against the author’s earlier benchmark, AARRI-Bench, whose equivalent prior-art comparison consumed effort second only to producing the benchmark data itself, for a table of just 12 rows and 7 columns.

**Table 1: Comparison of Relevant AI research benchmarks. AARRI-Bench simultaneously supports end-to-end research evaluation, fine-grained research process assessment, researcher quality and multi-harness evaluation.**

| Bench Name                          | End-to-End Tasks | Fine-Grained Eval | Researcher-Quality Eval | Data Generation  | Multi-Harness Eval | #Tasks    |
|-------------------------------------|------------------|-------------------|-------------------------|------------------|--------------------|-----------|
| MLE-Bench (Chan et al. 2025)        | ✗                | ✓                 | ✗                       | Transfer&Compose | ✓                  | 75        |
| MLGym-Bench (Nathani et al. 2025)   | ✗                | ✓                 | ✗                       | Automatic        | ✗                  | 13        |
| EXP-Bench (Kon et al. 2025)         | ✓                | ✗                 | ✗                       | Automatic        | ✓                  | 461       |
| ResearchCodeBench (Hua et al. 2026) | ✗                | ✓                 | ✗                       | Transfer&Compose | ✗                  | 212       |
| MLR-Bench (Chen et al. 2026)        | ✓                | ✗                 | ✗                       | Automatic        | ✓                  | 201       |
| PaperBench (Starace et al. 2025)    | ✗                | ✓                 | ✗                       | Transfer&Compose | ✗                  | 8316      |
| AstaBench (Bragg et al. 2025)       | ✓                | ✓                 | ✗                       | Transfer&Compose | ✓                  | 2400+     |
| InnovatorBench (Wu et al. 2025)     | ✓                | ✗                 | ✗                       | Transfer&Compose | ✗                  | 20        |
| AIRS-Bench (Lupidi et al. 2026)     | ✗                | ✓                 | ✗                       | Automatic        | ✓                  | 20        |
| COMPOSITE-Stem (Waters et al. 2026) | ✓                | ✓                 | ✗                       | Manual           | ✗                  | 70        |
| ScienceBoard (Sun et al. 2025)      | ✗                | ✓                 | ✗                       | Manual           | ✗                  | 169       |
| <b>AARRI-Bench (Ours)</b>           | ✓                | ✓                 | ✓                       | <b>Manual</b>    | ✓                  | <b>82</b> |

Figure 6. The manually built 12-by-7 prior-art table for AARRI-Bench, the workflow that Radar now shortens.

Limits: Radar search is deterministic lexical matching rather than semantic retrieval, so a differently worded query can change the candidate set. The session used the Radar CLI together with the agent's general web search, so it does not isolate Radar alone. Repository and dataset availability is a snapshot, not a permanent label. The case documents one contributor's workflow; it is not a measured user study. Full evidence, including the summary table and session screenshots, is public in issue #492.

Contributor. Jiayu Wang, Xi'an Jiaotong University. Case and evidence: [github.com/ktwu01/benchmark-radar/issues/492](https://github.com/ktwu01/benchmark-radar/issues/492)

### Use it

Cite Benchmark Radar for source-linked benchmark discovery, catalog search, and vendor-reporting evidence. Use the source and protocol fields when you compare records or scores.

## 7. Reproducibility, access, and citation

This report evaluates Benchmark Radar v0.9.0 at Git commit 98c7de3 and data cutoff 2026-08-29. The clean worktree ran the CI sequence: lint and formatting checks, external normalization, KW-Bench classification, checksummed data-release construction, and the full test suite. The current full CI suite passed.

### 7.1 Methods and limitations

The 6.2 audit uses the curated score archive and model-card registry. Raw headroom is the distance from one reported value to the metric bound. Repeat-controlled evidence requires the same instrument and protocol across at least two dates. Two dates support only a paired comparison. Because each repeated setup comes from one organization, the audit cannot estimate a field-wide trend. Sensitivity tables keep unknowns out of the eligible denominator.

### 7.2 Contributor credit

Junjie Zhou prepared the protocol audit, machine-readable table, and report revision for issue #457.

### 7.3 Issue link

Issue #457 Near-ceiling metrics under protocol controls.

| Artifact          | Canonical or permanent location                                                                                                                 |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Technical report  | <a href="https://doi.org/10.5281/zenodo.22167102">https://doi.org/10.5281/zenodo.22167102</a>                                                   |
| Source code       | <a href="https://github.com/ktwu01/benchmark-radar">https://github.com/ktwu01/benchmark-radar</a>                                               |
| Dashboard         | <a href="https://benchmark-radar.org/">https://benchmark-radar.org/</a>                                                                         |
| Cumulative JSON   | <a href="https://benchmark-radar.org/data/radar.json">https://benchmark-radar.org/data/radar.json</a>                                           |
| Benchmark catalog | <a href="https://benchmark-radar.org/data/benchmark-index.json">https://benchmark-radar.org/data/benchmark-index.json</a>                       |
| RSS               | <a href="https://benchmark-radar.org/feed.xml">https://benchmark-radar.org/feed.xml</a>                                                         |
| Citation metadata | <a href="https://github.com/ktwu01/benchmark-radar/blob/main/CITATION.cff">https://github.com/ktwu01/benchmark-radar/blob/main/CITATION.cff</a> |

### Published v0.9.0 citation

Wu, K. (2026). Benchmark Radar v0.9.0: Technical Report. Zenodo. <https://doi.org/10.5281/zenodo.22167102>

### Data statement

The v0.9.0 core and frozen section 3 rows use commit 98c7de3 and cutoff 2026-08-29. The issue #456 addendum separately audits 37 documents, 12 organization labels, and 110 benchmark IDs from a SHA-256-pinned registry at commit 98c8cf6 and cutoff 2026-08-31. Do not combine these populations; cite rolling dashboard values with a retrieval date.

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